Senior Software Engineer Interview Preparation

Candidate: Rohit Jangir S

Tell Me About Yourself (Full Version)

“Sure — I’m Rohit Jangir, a Software Engineer based in Bangalore with a little over 3 years of experience in backend and cloud-native application development. I’ve been working at Nagravision India Pvt Ltd, where I design and develop scalable microservices using Java, Spring Boot, and AWS. One of my key projects involved building a data migration service that processed over 15,000 accounts per hour for more than 50,000 users — improving integration reliability and system performance by around 20%. I also introduced a multi-tenant architecture that allowed us to host multiple customers on shared AWS infrastructure, significantly reducing our cloud costs. Along the way, I worked extensively with RabbitMQ for asynchronous communication and Docker and Kubernetes for containerized deployments, which improved our release speed by 30%. I really enjoy solving performance and scalability challenges, and I’m passionate about writing clean, maintainable code. I also like mentoring juniors and collaborating on architectural decisions. At this stage in my career, I’m looking for a role where I can contribute to system design and architecture decisions, while continuing to grow my technical depth — especially in distributed systems and cloud technologies — and that’s what attracted me to this opportunity.”

Tell Me About Yourself (Short Version)

“Hi, I’m Rohit Jangir from Bangalore. I have over 3 years of experience as a backend engineer, mainly using Java, Spring Boot, and AWS. At Nagravision, I’ve built and optimized scalable microservices, implemented a multi-tenant architecture to cut cloud costs, and improved system performance by about 20%. I also work with RabbitMQ, Docker, and Kubernetes for distributed deployments. I really enjoy solving performance and scalability challenges, and I’m now looking to take on a Senior Software Engineer role where I can contribute to system design and cloud architecture while mentoring others.”

1. Technical Depth & Problem Solving

Can you walk me through a complex system or project you've designed recently?

At Nagravision, I designed and developed a set of microservices using Java and Spring Boot for our content delivery platform. The services handled data migration for over 50,000 users, processing around 15,000 accounts per hour. I built asynchronous communication using RabbitMQ, containerized with Docker, and deployed on Kubernetes, improving performance by 20% and deployment time by 30%.

What was the most technically challenging problem you solved?

We had performance issues during large data migrations. I implemented batch inserts, optimized indexing in MongoDB/PostgreSQL, and used parallel streams in Java, improving migration speed 3x and reducing CPU load by 25%.

How do you ensure code quality and maintainability?

I follow clean code practices, use unit and integration tests with JUnit/Mockito, and enforce code reviews via Jenkins CI/CD. I introduced SonarQube to monitor coverage and reduce tech debt.

How do you approach debugging production issues?

I check AWS CloudWatch logs and metrics, trace with correlation IDs, reproduce locally, validate with tests, and deploy fixes via Jenkins. I also document the root cause in Confluence.

What's your process for reviewing others' code?

I focus on logic correctness, architecture fit, and readability. I provide constructive comments, verify test coverage, and ensure adherence to team coding standards.

2. System Design & Architecture

Design a scalable API/service for file uploads.

I'd design a Spring Boot service generating pre-signed S3 URLs, store metadata in PostgreSQL, process uploads via RabbitMQ workers, and deploy on Kubernetes with autoscaling.

How would you design a system to handle millions of concurrent requests?

I'd use stateless Spring Boot services behind AWS ALB, cache with Redis, shard databases, serve static assets via CloudFront, and apply circuit breakers and centralized logging.

What design principles do you follow?

SOLID, separation of concerns, and observability-first design with logging and tracing from day one.

How do you choose between databases or messaging systems?

I use PostgreSQL for relational data, MongoDB for flexible data, Neo4j for graphs. RabbitMQ for reliable queues, Kafka for high-throughput streams.

3. Teamwork & Leadership

How do you mentor junior engineers?

I pair program, explain design reasoning, and encourage ownership through small feature delivery.

Tell me about a time you influenced a technical decision.

I led a POC comparing microservices vs. monolith, demonstrating scalability benefits that influenced company-wide adoption.

How do you handle disagreements during design reviews?

I rely on data, benchmarks, and team discussion. If unresolved, we run small experiments to decide objectively.

How do you balance delivery speed and technical debt?

I track debt in Jira, dedicate 10–15% of sprint capacity to fix it, ensuring long-term quality.

4. Ownership & Impact

Tell me about a time you took ownership beyond your scope.

I investigated and fixed a production Nginx issue outside my module, restoring service and earning a Bravo Award.

What's a project where you had measurable business impact?

Implemented multi-tenant architecture reducing AWS costs and improving scalability without losing isolation.

How do you ensure features meet business needs?

I clarify requirements with PMs, track post-release metrics, and perform impact validation after deployment.

5. Motivation & Fit

Why are you interested in this role?

This role aligns with my experience in backend systems, Java, and AWS. I'm excited to contribute to design discussions and scalable architectures.

What are you looking for in your next opportunity?

A chance to take architectural responsibility, mentor others, and deepen cloud/distributed systems expertise.

How do you approach learning new technologies?

I build POCs and follow documentation. Recently explored Rust and AWS ML APIs via Udacity.

What kind of environment helps you perform best?

Collaborative teams with open communication, ownership, and focus on engineering excellence.

Good Questions to Ask the Hiring Manager

- What does success look like for a senior engineer in your team?
- How does your team handle technical design decisions?
- What are the biggest technical challenges you're currently solving?
- How do you encourage continuous learning or innovation?